

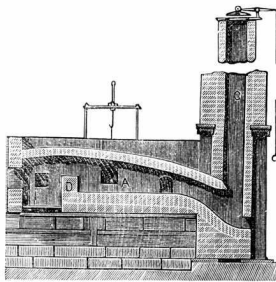
# HENRY CORT & THE FUNTLEY IRONWORKS (1783–1784)

Supporting the Hampshire Archives Trust Competition & Year 8 Industrial Revolution Enquiry

**CORE ENQUIRY QUESTION:** "To what extent was Henry Cort's breakthrough at Funtley the result of technological innovation versus geographic proximity to Portsmouth Royal Dockyard?"

## DOCUMENT 1: PATENT SPECIFICATION NO. 1420

1784 · TECHNICAL DIAGRAM

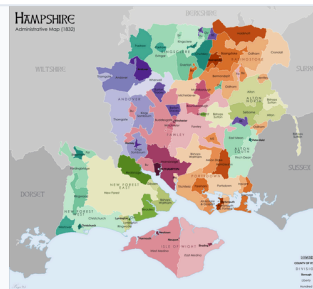


"My new method of manufacturing iron in a reverberatory air furnace heated by common raw pit coal... and rolling the same between grooved rollers without repeated hammering."

**Archival Significance:** Cort's diagram proves the coal fuel was separated from the iron ore. The heat bounced off the curved ceiling, burning off carbon without contaminating the iron with sulfur.

## DOCUMENT 2: RIVER MEON SURVEY & ROUTE

C. 1780S · CARTOGRAPHIC SURVEY



"Funtley Mill situated upon the Meon Stream, with water pond and millrace driving twin waterwheels... 2 miles from Fareham Haven Quay and 9 miles to Portsmouth Harbour."

**Archival Significance:** Before steam engines were installed, Cort was reliant on the natural watercourse of the River Meon to turn the massive iron rollers, with easy river transport down to the Solent.

## DOCUMENT 3: ADMIRALTY BOARD & NAVY BOARD TRIAL MINUTES

PORTSMOUTH DOCKYARD · MARCH 1784 · OFFICIAL STATE REPORT

"Pursuant to your Lordships' directions, trials have been made at Portsmouth Dockyard of the malleable iron manufactured by Mr Henry Cort at Fontley near Fareham. The master smiths report that Cort's iron was subjected to severe bending, cold-hammering, and welding into ship knees, mast hoops, and anchor shackles. They unanimously declare it to exceed in toughness and tensile strength any iron made in this kingdom, and to be fully equal to the finest imported Swedish Orgrounds iron."

**Archival Analysis:** Prior to 1784, Britain imported over 80% of its naval wrought iron from Sweden and Russia. This official Admiralty verification established Cort's Funtley works as a national strategic asset, transforming Portsmouth into the world's most formidable naval base.

## YOUNG HISTORIAN INVESTIGATION PROMPTS (HAMPSHIRE ARCHIVES TRUST GUIDE)

### 1. PROVENANCE & MOTIVE

Why was Henry Cort, an ex-navy agent, uniquely motivated to patent a process using cheap 'pit coal' instead of scarce timber charcoal?

### 2. ARCHIVAL CORROBORATION

How does Document 3 (Portsmouth Dockyard trial) corroborate the mechanical claims Cort made in Document 1 (his patent)?

### 3. LOCAL VS NATIONAL IMPACT

Could Cort's breakthrough have happened without the water rights of the River Meon and the customer base of Portsmouth?

